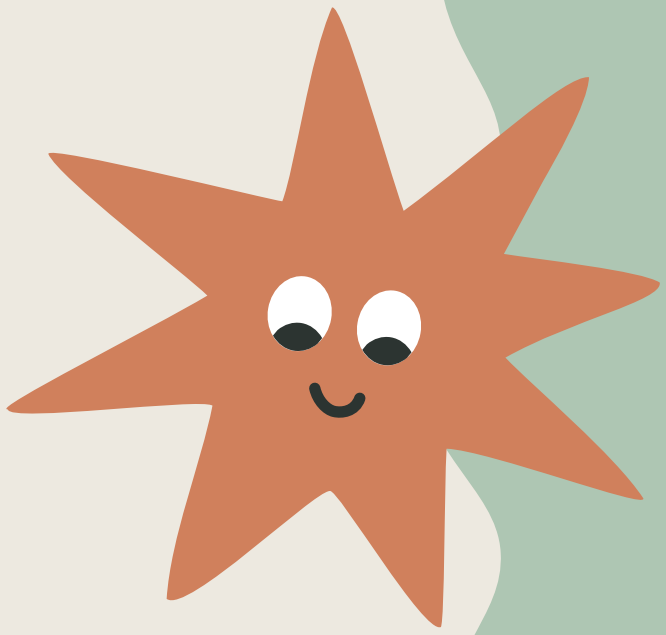
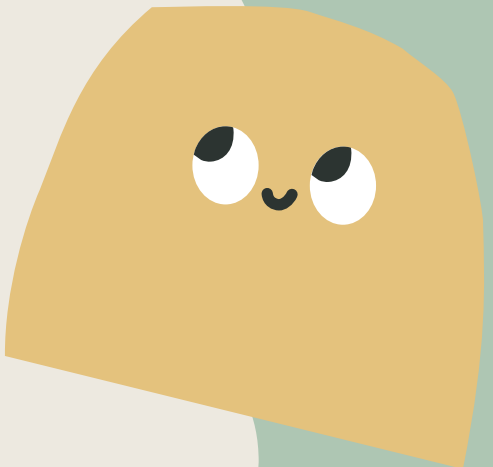
Strings

Lecture 4





Learning Objectives

➢By the end of this lecture, students should be able to:

- Understand what a string is.
- Create strings in Python.
- Use single and double quotes.
- Access individual characters using indexing.
- Understand positive and negative indexing.
- Extract portions of strings using slicing.
- Use common string methods.
- Format strings using f-strings.
- Build practical programs involving text.

Prepared By: Ehtisham Rasheed

What is a String?

- A **string** is a sequence of characters
- A string can contain letters, numbers, spaces, symbols
- String with quotes
- Multi strings
- String indexing
- Negative indexing
- Invalid index

String:	P	Y	T	H	O	N
Positive:	0	1	2	3	4	5
Negative:	-6	-5	-4	-3	-2	-1

Prepared By: Ehtisham Rasheed

String Slicing

- Indexing gives us **one character**
- Slicing allows us to extract **multiple characters**
- Syntax

```
string[start:end]
```

  - The start index is included
  - The end index is not included
- Slicing to the end
- Negative slicing
- Step in slicing

```
string[start:end:step]
```

- Reversing a string

Prepared By: Ehtisham Rasheed

More on Strings

- Strings are immutable
- Common string methods
  - Upper
  - Lower
  - Capitalize
  - Title
  - Strip
  - Replace
  - Find
  - Count

Prepared By: Ehtisham Rasheed

More on Strings

- Common string methods
  - startsWith and endsWith
  - split and join
  - isalpha, isdigit and isalnum
  - f-Strings
  - Expressions inside f-String
  - Formatting numbers with f-String
  - len() function

Prepared By: Ehtisham Rasheed